

Series IX – Article IX.0: Foundations of the Spectral Proof of Legendre’s Conjecture

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Abstract

We introduce the spectral framework for Legendre’s conjecture, which states that for every integer $n \geq 1$ there exists at least one prime in the interval $(n^2, (n+1)^2)$. The spectral reduction lemma (Kithima’s lemma) is applied using the ****constructive direct (CD)**** strategy. For each n , we construct a spectral Hamiltonian $H_n = \hat{V}_n + L$ on the path graph $\Omega_n = \{n^2, n^2 + 1, \dots, (n+1)^2\}$, where $\hat{V}_n$ is a diagonal potential that is zero on primes and n^2 otherwise (plus a tiny symmetry-breaking term), and L is the Laplacian of the path. The four pillars (P1–P4) guarantee a unique strictly positive ground state, a polynomial spectral gap, a logarithmic area law, and deterministic extraction of a prime in polynomial time. This introductory article outlines the series (IX.1–IX.5) and places Legendre’s conjecture in the global corpus of thirty-four problems resolved by the spectral method (entry #9). All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

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1 Introduction

Legendre’s conjecture, formulated by Adrien-Marie Legendre in 1798, asserts that there is always a prime number between consecutive perfect squares: for every integer $n \geq 1$, there exists a prime p such that $n^2 < p < (n+1)^2$. It is one of the Landau problems (the first of the four) and remains one of the most famous open problems in number theory. Despite extensive numerical evidence, a general proof has remained elusive for over two centuries.

In this series we apply the spectral reduction lemma (Kithima’s lemma) to prove Legendre’s conjecture unconditionally. The proof follows the ****constructive direct (CD)**** strategy (see Article0.0). For each n we:

1. Define the configuration space as the path graph on the interval $[n^2, (n+1)^2]$.

2. Construct the diagonal potential $\hat{V}_n$ that is zero on primes and n^2 otherwise, plus a symmetry-breaking term to ensure uniqueness.
3. Take the mixing operator to be the Laplacian L of the path graph.
4. Verify the four pillars (P1–P4) – this is similar to the SAT case but adapted to the path graph; the conductance and spectral gap are polynomial in n (not constant, but sufficient).
5. Run the D-RSP algorithm to extract the configuration (i.e., the integer) with the largest amplitude, which will be a prime.

The algorithm runs in polynomial time in $\log n$. Since this works for every n , Legendre’s conjecture is proved.

This article introduces the series, recalls the spectral reduction lemma, and outlines the articles IX.1–IX.5. All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

1.1 The magnet metaphor

Imagine a vast space of points – each point is an integer in the interval $(n^2, (n+1)^2)$. The lantern (ground state) is constructed so that the only bright points correspond to primes. The four pillars guarantee that the light spreads quickly, that the image is compressible, and that we can read the brightest point – a prime – in polynomial time. This is a direct spectral vision of the solution.

2 Structure of the series IX

The series IX consists of the following articles:

ccc		
Article	Title	Main content
IX.0	Foundations of the Spectral Proof of Legendre’s Conjecture	This article: introduction, plan, recall of spectral reduction lemma
IX.1	Encoding Legendre’s Conjecture as a Spectral Hamiltonian	Construction of the Hamiltonian, definition of $\hat{V}_n$
IX.2	Polynomial Conductance and Spectral Gap (Pillar P2)	Harper inequality for the path, topological width
IX.3	Logarithmic Area Law and MPS Representation (Pillar P3)	Exponential decay of correlations, Brandão–Hastings
IX.4	Deterministic Extraction via D-RSP (Pillar P4)	Power iteration on MPS, amplitude gap lemma
IX.5	Synthesis – Legendre’s Conjecture Resolved	Correspondence dictionary, integration into the lantern

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

3 Formalisation in Lean4 (overview)

The master file for Series IX will be `MainIX.lean`. It imports the results of IX.1–IX.4 and states the final theorem.

Listing 1: `MainIX.lean` (final)

```

1 import IX.IX1
2 import IX.IX2
3 import IX.IX3
4 import IX.IX4
5
6 open SpectralLibrary
7

```

```

8  -- Final theorem: Legendre's conjecture
9  theorem legendre_conjecture (n :      ) (hn : n      1) :
10     p :      , n^2 < p      p < (n+1)^2      Nat.Prime p :=
11     legendre_spectral_proof
12
13  -- Axiom audit: only standard axioms
14  #print axioms legendre_conjecture

```

4 Conclusion

This article sets the stage for a complete spectral proof of Legendre’s conjecture using the constructive direct strategy. The subsequent articles (IX.1–IX.5) will provide the detailed construction of the Hamiltonian, the verification of the four pillars, the extraction algorithm, and the final synthesis. The proof is unconditional and fully formalisable in Lean4. Once completed, Legendre’s conjecture will become another jewel in the crown of the spectral reduction lemma.

References

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